

02 – Chemical Language and Nomenclature

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1 Elements and Chemical Symbols

1.1 The periodic table

An *element* is a substance whose atoms all share the same number of protons. The periodic table orders elements by increasing atomic number Z , arranged so that elements with similar valence configurations fall in the same column (group).

The recurring chemical behavior down a group is a direct consequence of the shell structure derived in Chapter 01: elements in the same group share the same valence electron configuration (e.g. ns^1 for the alkali metals), so periodic trends in atomic radius, ionization energy, and electronegativity are simply different faces of the same underlying orbital filling pattern (1s, 2s, 2p, 3s, ...) dictated by the Aufbau principle and the Pauli exclusion principle.

1.2 Atomic number

The *atomic number* Z is the number of protons in the nucleus. It uniquely identifies the element and equals the number of electrons in a neutral atom.

The electronic structure of an atom follows from the nuclear charge $+Ze$ acting on the electrons through the Coulomb potential

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}, \quad (1.1)$$

which enters the electronic Schrödinger equation discussed in Chapter 1. Changing Z changes the element; changing the electron count while Z is fixed only changes the ionization state.

1.3 Atomic mass

The *atomic mass unit* (u, or dalton, Da) is defined as 1/12 the mass of a ^{12}C atom:

$$1\text{ u} = \frac{1}{12} m(^{12}\text{C}) \approx 1.6605 \times 10^{-27} \text{ kg}. \quad (1.2)$$

The *atomic mass* (or relative atomic mass, A_r) listed on the periodic table is the isotopic-abundance-weighted average

$$A_r = \sum_i f_i m_i, \quad (1.3)$$

where f_i is the natural fractional abundance of isotope i and m_i its isotopic mass in u.

Worked Example 1.1: average atomic mass of chlorine

Chlorine occurs naturally as ^{35}Cl ($m = 34.969\text{ u}$, $f = 0.7576$) and ^{37}Cl ($m = 36.966\text{ u}$, $f = 0.2424$). Then

$$A_r(\text{Cl}) = (0.7576)(34.969) + (0.2424)(36.966) = 35.45\text{ u}, \quad (1.4)$$

matching the tabulated value. Note that no single chlorine atom actually has this mass – it is a statistical average over the isotopic population.

1.4 Chemical symbols

Each element is denoted by a one- or two-letter symbol (e.g. H, He, Na, Fe), standardized by IUPAC. Many symbols derive from Latin or Greek names (Na from *natrium*, Fe from *ferrum*, W from *wolfram*) rather than the modern English name.

1.5 Isotopic notation

Isotopes of an element share the same Z but differ in neutron number N , hence in mass number

$$A = Z + N. \quad (1.5)$$

The standard notation is

$${}^A_Z\text{X} \quad (1.6)$$

e.g. ${}^2_1\text{H}$ (deuterium) or ${}^{13}_6\text{C}$.

2 Chemical Formulas

2.1 Empirical formula

The *empirical formula* gives the simplest whole-number ratio of atoms of each element in a compound. It is obtained by dividing the moles of each element by the smallest mole value among them.

Worked Example 2.1: Empirical formula from mass composition

A compound is found to contain 40.0% C, 6.7% H, and 53.3% O by mass. Taking 100 g as basis,

$$n_{\text{C}} = \frac{40.0}{12.01} = 3.33\text{ mol}, \quad n_{\text{H}} = \frac{6.7}{1.008} = 6.65\text{ mol}, \quad n_{\text{O}} = \frac{53.3}{16.00} = 3.33\text{ mol}. \quad (2.1)$$

Dividing by the smallest value (3.33) gives the ratio C : H : O = 1 : 2 : 1, so the empirical formula

is CH_2O .